Englewood Rd. | Heating System Master Reference

Living technical inventory, wiring map, operating model, diagnostic record & open questions

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How to use this document

Evidence tags: [CONFIRMED] directly reported or measured by homeowner; [OBSERVED] visible in photos or equipment labeling; [DOCUMENTED] supported by an attached manufacturer document; [HYPOTHESIS] plausible but not traced; [OPEN] unresolved. A confirmed report is not the same as an independently verified wiring trace. Never treat a hypothesized signal path as as-built wiring.

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1. System at a glance

- [CONFIRMED] Two Polaris water heaters in series supply hydronic space heating; the system also serves domestic hot-water functions, but the complete domestic-water routing is not yet mapped.
- [CONFIRMED] Six installed radiant zones: Basement; Living Area; Master Bedroom; Master Bath; Rest of Second Floor; Bedroom Above Garage (“Yellow Room”).
- [CONFIRMED] Two Lennox AM30 hydronic air handlers: one in basement serving basement/main floor/master areas; one in attic serving upstairs areas.
- [CONFIRMED] Five air thermostats use first-stage radiant and second-stage supplemental forced air, as described by homeowner; exact staging and call retention are unverified.
- [OBSERVED] Two three-channel Wirsbo zone-controller boards, external black relay assemblies, a MARS E2-4 four-zone component, and a Wirsbo DuoMix 201 injection mixing control.
- [CONFIRMED] Basement radiant is separate from the DuoMix and uses a Caleffi 521 thermostatic mixing valve and dedicated circulator/relay.
- [CONFIRMED] Original proposed schematic included Garage Entry zone; homeowner says it was never installed.

2. Heating zones and equipment mapping

Zone	Wirsbo channel	Hydronic grouping	Supplemental air / notes
Basement	Independent control (not one of two 3-zone boards)	Separate Caleffi 521 mixing + circulator	Basement AM30; detailed thermostat/relay path open
Living Area	Upper board, left / zone 1	DuoMix Mix 1 (reported)	Basement AM30
Master Bedroom	Upper board, center / zone 2	DuoMix Mix 1 (reported)	Basement AM30
Master Bath	Upper board, right / zone 3	DuoMix Mix 1 (reported)	Floor sensor; air-stage behavior needs verification
Rest of Second Floor	Lower board, left / zone 1	DuoMix Mix 2 (reported)	Attic AM30; current fault under investigation
Bedroom Above Garage / Yellow Room	Lower board, center / zone 2	DuoMix Mix 2 (reported)	Attic AM30; supply sensor cable discovery
Spare	Lower board, right / zone 3	Unused	Handwritten SPARE tag

[OPEN] Confirm the exact count and location of five thermostats against six radiant zones, especially the Master Bath floor-sensing control.

3. Hydronic generation and distribution

3.1 Polaris water heaters

[CONFIRMED] Two Polaris heaters are connected in series. Homeowner notes list model “PG10 34-150-2NV”; verify that this model applies to both units and capture serial numbers, years, venting, pumps, setpoints and any interlocks. Existing parts notes include condensate trap 186642-000, AS568-327 O-ring, and stainless washer dimensions; these are reference notes, not proof of installed parts.

3.2 Radiant mixing and pumps

- [CONFIRMED] Basement: Caleffi 521 thermostatic mixing valve; independent circulator controlled by relay.
- [OBSERVED] DuoMix 201 handles two mixed-water groups and has outputs labeled Mix 1 Pmp, Var 1 Pmp, Mix 2 Pmp, Var 2 Pmp.
- [CONFIRMED] Upstairs circulator identified in homeowner notes as Taco 008-BC6; parts compatibility noted as Taco 008-045RP cartridge; confirm exact pump nameplate before parts purchase.
- [OPEN] Location, make/model, power source, and hydraulic role of each Mix 1/Mix 2 and Var 1/Var 2 pump; manifold and valve locations; flow direction and check valves.

3.3 Lennox air handlers

[CONFIRMED] Basement and attic Lennox AM30 hydronic air modules provide supplemental forced-air heating. Attached Lennox HM30/AM30 engineering bulletin is a product-family reference, not

proof of the exact installed configuration. Capture actual unit nameplates, control wiring, and supply/return connections.

4. Controls and terminal reference

4.1 Wirsbo DuoMix 201

[OBSERVED] Faceplate: Wirsbo DuoMix 201, part A3030005, marked tekmar / March 1999. Do not conflate with Wirsbo proMix 201 or proMix 212. The attached tekmar SB 051 cross-reference is explicitly for proMix products, and does NOT establish an equivalent replacement for this DuoMix 201.

Terminals	Printed function	What is known
1–2	Mix 1 Demand	Input; source not yet traced
3–4	Mix 2 Demand	Input; crucial for current fault
5–6	Boiler Enable	Function labeled; connected equipment unknown
7 N / 8 Mix2 Pmp	Mix 2 circulation pump	No 120 V at terminal 8 during upstairs call (user measurement)
9 N / 10 Var2 Pmp	Mix 2 variable pump	Operating behavior not measured
11 N / 12 Mix1 Pmp	Mix 1 circulation pump	120 V measured at terminal 12 on first-floor call
13 N / 14 Var1 Pmp	Mix 1 variable pump	Operating behavior not measured
15 N / 16 L	120 V supply	Supply labeling visually identified
17 COM / 20 SUP 2	Supply temperature sensor 2	Brown/blue conductors from Yellow Room cable reportedly connect to pipe-mounted sensor
18–19, 21–27	Other sensor / control terminals	Capture exact labels and connections before documenting

[OBSERVED] Display-side legends include Mix 1 Demand, Mix 2 Demand, Minimum, Maximum, Melting, Idling and CWCO. Display icons / “OPEN” indications are not yet decoded against the manufacturer manual.

4.2 Wirsbo zone controllers

[OBSERVED] Two 3-zone Wirsbo boards. Board printing in recent photo appears “A303?0003”; exact part number must be read from a sharp straight-on photo. Each channel has top THERMOSTAT four-position terminals (visibly R1, R2, Y1, Y2), lower TELESTAT four-position terminals, and indicators. A separate right-edge OUTPUT terminal strip is present. Its circuit type and function are not yet verified.

[OBSERVED] On the lower board, a yellow light illuminates for Rest of Second Floor thermostat call. Green indicator near board power supply has been seen. These indicators do not, by themselves, prove that the OUTPUT terminals close or that the zone valve opens.

4.3 Other cabinet hardware

- [OBSERVED] Three black relay-like devices mounted above each Wirsbo board, apparently one per zone; function and terminal numbering need positive identification. The latest photo shows wiring from them to THERMOSTAT terminals.
- [REPORTED] MARS E2-4 four-zone board is also present. Its exact place in thermostat staging has not been traced.
- [OPEN] Confirm if Honeywell valves are controlled by these external relays, by the Wirsbo TELESTAT terminals, or by another interface.

5. Thermostats, relays, valves and cable routing

5.1 Cable discovery — upstairs

- [CONFIRMED] Multiconductor brown-sheathed low-voltage cable exits top of control cabinet; conductors in same jacket serve more than one function.
- [CONFIRMED] Bedroom Above Garage / Yellow Room: brown and blue conductors from a brown cable go to DuoMix terminals 17 COM and 20 SUP 2. A pipe-contact temperature sensor exists at the far end, per homeowner.
- [CONFIRMED] Rest of Second Floor: brown conductors from its cable run to external black relay devices, rather than to DuoMix SUP 2. Exact relay terminals and coil/contact functions are not yet traced.
- [OBSERVED] The black relays connect by wires to the Wirsbo THERMOSTAT terminal blocks. This makes an upstream/isolation relay role plausible but not proven.
- [CONFIRMED] Honeywell zone valves are installed instead of the Wirsbo Telestat-style thermal actuators shown in earlier generic examples. Their model and wiring arrangement are unknown.

5.2 Critical electrical distinction

[DOCUMENTED BY LABELS] THERMOSTAT is an input interface; TELESTAT is an actuator-oriented interface; OUTPUT is a distinct terminal strip. Whether the installed Honeywell motorized zone valves are powered directly by TELESTAT outputs or through other relays remains [OPEN]. Honeywell valves often have separate motor and end-switch circuits, but no specific installed model or end-switch wiring has been confirmed.

6. Operating sequence — verified vs proposed

6.1 Observed signal fragments

- [CONFIRMED] Upstairs thermostat call → lower Wirsbo board zone-1 yellow indicator illuminates.
- [CONFIRMED] Upstairs call → DuoMix display reportedly shows no visible change; Mix 2 pump hot output (terminal 8) measured inactive.
- [CONFIRMED] First-floor thermostat call → DuoMix Mix 1 pump hot output (terminal 12) measured at 120 V.
- [CONFIRMED] Yellow Room cable spare conductors → SUP 2 temperature sensor, not a demand input.

6.2 Unverified possible sequence

[HYPOTHESIS] Upstairs thermostat → external relay and/or Wirsbo THERMOSTAT input → zone-valve motor control → valve end switch or zone-board OUTPUT → DuoMix Mix 2 Demand (3–4) → Mix 2 circulator (7–8), with Supply 2 sensor modulating VAR 2 injection pump. Each arrow beyond the first observed thermostat/LED relationship needs a continuity trace or safe functional test. Some systems signal demand independently of valve end switches.

7. Troubleshooting log: second-floor circulator

Observation	Interpretation / limitation
Rest of Second Floor thermostat calls	Initiating request is present
Lower board zone 1 yellow indicator lights	Board registers or indicates some zone activity; LED exact meaning unverified
No visible DuoMix LCD response	Possible missing Mix 2 demand, or display behavior not understood
No 120 V at DuoMix terminal 8	Mix 2 pump output not active at test moment; measurement reference and conditions should be recorded
120 V at terminal 12 with first-floor call	Mix 1 output demonstrably works under tested condition
Historic note: “no output on zone 3”	Past board issue suspected; no evidence that current zone 1 has same defect

7.1 Leading hypotheses — not diagnoses

- H1: Mix 2 demand signal never reaches DuoMix terminals 3–4 due to wiring, board OUTPUT, relay, or end-switch problem.
- H2: A Honeywell valve fails to open or close its end switch, blocking a demand chain if wired that way.
- H3: DuoMix receives demand but inhibits Mix 2 circulation because of a setting, sensor condition, or internal fault.
- H4: External relay or Wirsbo zone-controller defect; historic zone-3 note raises interest but is not direct evidence.
- H5: Measurement or timing issue (e.g., valve opening delay, demand timeout, or misunderstanding of output logic).

7.2 Next discriminating checks

- Trace the actual two wires attached to DuoMix 3–4 to their origin, without assuming they originate at the Wirsbo OUTPUT strip.
- Photograph and identify terminal labels and model numbers of the black relays; determine which terminals are coil and which are contacts.
- Photograph Honeywell valve model and terminal/wire colors, then document whether valve moves during a call and whether end switch is wired.
- Document what lower Wirsbo right-side OUTPUT terminals do with and without a call, using manufacturer wiring instructions and qualified low-voltage testing.
- Observe DuoMix Mix 2 Demand indicator with a sustained call, and verify controller settings and sensor readings.

SAFETY: Cabinet includes 120 VAC and 24 VAC. Do not bridge demand terminals, move wires, or probe live terminals unless their ratings and circuit isolation are established and the person testing is qualified. Never use a neutral terminal as a substitute for a verified meter reference.

8. Historical service evidence

[CONFIRMED] A box for a replacement Wirsbo zone controller contains a handwritten note “no output on zone 3.” The identity of the board, the date, the author, whether it was replaced, and which “output” was meant are unknown. This is evidence of a prior reported fault, not evidence of current zone-1 failure. Photograph box, labels, note, and any board inside.

9. Priority knowledge gaps and investigation plan

Priority	Question	Why it matters
P0	Where do the two conductors on DuoMix Mix 2 Demand terminals 3–4 physically terminate?	Directly establishes whether demand path is intact.
P0	What are the model, pinout, and coil/contact wiring of the external black relays above each Wirsbo board?	Resolves upstream vs downstream role and may identify failed stage.
P0	What is the exact model and wiring of each Honeywell zone valve? Do they have end switches, and where do those switches go?	Determines whether valve position gates pump demand.
P0	What do the right-side Wirsbo OUTPUT terminals electrically represent and which wires leave them?	Tests leading theory of missing Mix 2 demand.
P1	Does Mix 2 Demand at DuoMix 3–4 actually change state during a sustained upstairs call?	Separates upstream control failure from DuoMix behavior.
P1	Does Rest of Second Floor valve physically open on a call, and how long does it take?	Separates valve actuation from pump command.
P1	Where does each thermostat cable terminate first: MARS, external relay, or Wirsbo board? What are Y1/Y2/R1/R2 used for?	Documents stage 1/2 signal topology.
P1	What is the exact Wirsbo board part number and its manufacturer wiring diagram?	Allows proper interpretation of LEDs and outputs.
P1	What is the exact DuoMix 201 installation/service manual, and what do its LCD states and pump interlocks mean?	Prevents misreading demand and lockout behavior.
P1	Are Mix 2 and Var 2 separate physical pumps? Identify and photograph each pump and its wiring.	Distinguishes circulation from injection control.
P2	Which of the five air thermostats serves each of the six radiant zones? Is Master Bath a separate floor-sensor	Completes zone-to-thermostat mapping.

	control?	
P2	Does stage 2 forced air keep stage 1 radiant active or drop it? How is AM30 staging implemented?	Explains simultaneous calls and relay complexity.
P2	Where are the manifolds and all Honeywell zone valves? Which loops are on each?	Completes hydraulic map.
P2	What are the precise model/serial numbers, age, setpoints and service status of both Polaris heaters and both AM30s?	Enables maintenance and replacement planning.
P2	What was replaced historically when “no output on zone 3” was written, and which board was affected?	Determines relevance of historical fault.
P2	What are the DuoMix sensor placements (SUP 1, SUP 2, outdoor, return) and calibration/readings?	Verifies mixing control inputs.
P3	Where are isolation valves, purge points, expansion tank, pressure/temperature gauges and safety controls?	Supports maintenance and safe diagnosis.
P3	What is the actual as-built difference from the original proposed piping schematic?	Preserves accurate as-built record.

Recommended field capture format

For every cable: assign ID, jacket color, origin, destination, conductor color, terminal number, voltage/type, and photo reference. For every test: date/time, thermostat setpoint, demand duration, LEDs/display, valve position, meter points and readings. Preserve a photo before moving any wiring.

10. Documents, photos and reference inventory

- Info-on-water-heaters.txt — homeowner Polaris model, service parts and reference URLs.
- pasted.txt — homeowner notes on radiant controls, Caleffi 521, contractors, upstairs Taco pump, and an unresolved proMix/tekmar cross-reference.
- sb0051_01.PDF — tekmar SB 051 (2008) pro Series cross-reference; covers proMix 201 and 212, NOT verified DuoMix 201.
- Taco-008-045RP-Replacement-Parts-List.pdf — Taco circulator parts family reference.
- lennox_am30_completeheat.pdf — Lennox HM30/AM30 engineering data, September 2002.
- IMG_28E85D4A-0A4B-4C2A-A624-58928E397ADB.jpeg — lower Wirsbo board with three external relay-like devices and thermostat terminal blocks.
- Other uploaded cabinet and control images: IMG_201F..., IMG_3BA..., IMG_4B3D..., IMG_6121..., IMG_649A..., IMG_7022..., IMG_B6A..., IMG_CD7...; associate each with caption during photo audit.
- Original proposed piping schematic was discussed but has not been unambiguously identified among current source files; attach and cite its exact filename in next revision.

11. Change log / future field notes

Date	Revision	Change
2026-11-07	v0.1	Initial master reference assembled from conversation observations, recent photo and uploaded reference files; questions prioritized.

Next revision: add photographed as-built terminal-by-terminal cable schedule, validated schematic, and closed/open status for each P0 question.